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Editors
Physical Review B

Dear Editors,

I submit the manuscript "Local-resolvent formulation and assessment of linear-scaling Kohn–Sham calculations" for consideration in Physical Review B.

The manuscript writes the Kohn–Sham density in terms of the diagonal of the one-electron resolvent sampled at a few complex energies and studies one simple realization of this idea. A half-step trapezoid rule on a circle gives an occupation function that always lies between 0 and 1. We prove an a priori error bound for this rule and show that, with a circle that moves with the chemical potential, forces at fixed electron number are exact at every number of poles.

The manuscript also reports where this realization falls short. Gauss–Legendre and Zolotarev pole rules need fewer poles, and the cost of the half-step rule is set by the ratio of occupied band width to gap. In three dimensions the method is a divide-and-conquer scheme. We compare it with divide-and-conquer using Fermi–Dirac occupations, which turns out to be equally accurate, and with dense and warm-started iterative Kohn–Sham solvers in the same basis. For gapped H_2 molecular solids it reproduces self-consistent Kohn–Sham energies to 2.5×10^{-7} Ha per electron up to 1024 electrons, and, because of a large prefactor, complete calculations are slower than single-threaded dense diagonalization below about 500 electrons (by up to a factor of six at 54 electrons), 2.0 times faster at 686 electrons and 3.5 times faster at 1024 electrons (three interleaved repeats on an idle machine; no comparison with optimized or parallel dense solvers). The cases in which the method is inaccurate (disorder, compression, small gaps and diffuse basis sets) are measured, and the relation to Green's-function, pole-expansion and divide-and-conquer methods is discussed in the text.

All analytic identities were checked by computer algebra, and the implementation is covered by automated tests. The Supplemental Material contains the solver comparisons, timing, memory and thread data, the basis-set, margin and pole-rule studies, window diagnostics and SCF convergence histories. The code and data are available at `https://github.com/albert-hzbn/LResDFT`. The manuscript is not under consideration elsewhere.

Sincerely,

Albert Linda